Recovered, but not back on track

World life expectancy at birth fell 1.7 years in the pandemic and was back above its 2019 level by 2022. In 2023 it stood at 73.2 years, still 1.1 years short of where the trend of 2000–2019 pointed.

Data
Our World in Data, 1950–2023; Natural Earth
Made with
HTML, matplotlib and `vellum.py`

Summary

1. **The situation.** World life expectancy rose 26.8 years since 1950; the pandemic took 1.7 back. Figure 1
2. **The trend.** In 2023 the world was 1.1 years below its trend, though above its 2019 level. Figure 2
3. **The gap.** Of the 35 countries with the lowest life expectancy, 31 are in Africa. Mexico, Colombia and South Africa fell four years or more and were back by 2023. Figures 3 and 4
4. **The options.** Three examples: measure recovery against the trend, look where the fall was deepest, keep the income gap in view.
5. **The ask.** An example: report life expectancy beside its pre-2020 trend every year until the gap closes.

73.2
years: world life expectancy at birth in 2023, 26.8 more than in 1950

–1.7
years: the world's fall from 2019 to 2021, the largest since 1960

1.1
years below the trend of 2000–2019 in 2023

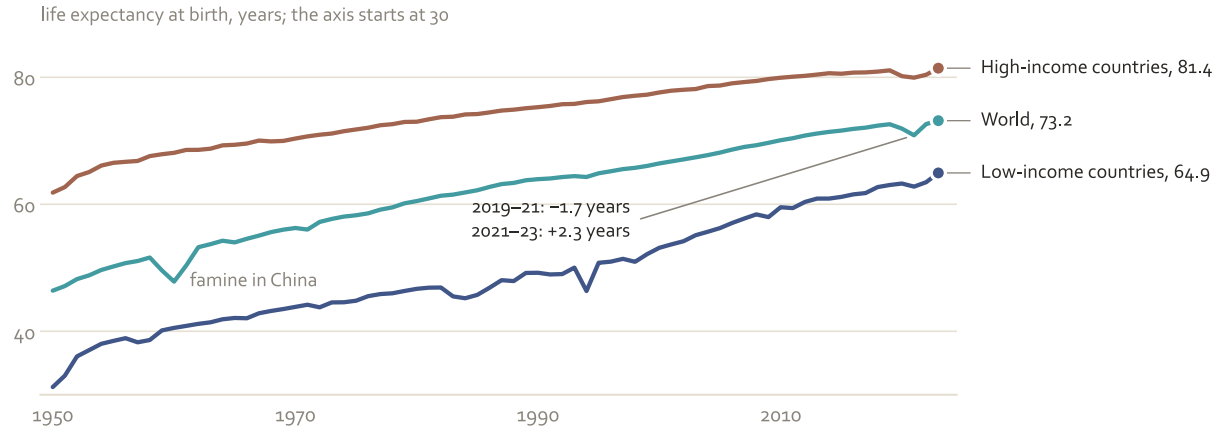
1 The situation

In 1950 a newborn could expect to live 46.4 years at that year's death rates; in 2023, 73.2. That is a gain of 26.8 years, 0.37 for every calendar year, and the gap between high- and low-income countries shrank from 30.6 to 16.5 years. Two falls break the world line: famine in China around 1960, when life expectancy there dropped from 48.8 to 33.4, and the pandemic, which took 1.7 years between 2019 and 2021.

How to read it. Period life expectancy at birth is the average number of years a newborn would live if the death rates of that year held for its whole life. A fall measures that year's deaths; it is not a forecast for the children born then.

World life expectancy rose 26.8 years since 1950; the pandemic took 1.7 back

Period life expectancy at birth, 1950–2023: the world, high-income countries and low-income countries



Our World in Data (UN WPP 2024 and others), 1950–2023 · make_report.py · git 4c16514

Figure 1. Income groups as defined in UN World Population Prospects. Source: UN World Population Prospects (2024) and others, via Our World in Data (CC BY 4.0); data for 1950–2023.

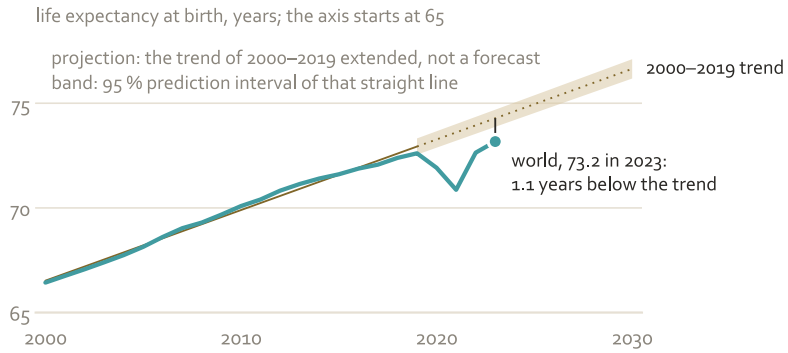
2 The trend

By 2022 the world was back above its 2019 level. Against its trend it has not recovered: from 2000 to 2019 life expectancy rose 3.4 years a decade, and that straight line points to 74.3 years in 2023. The world was 1.1 years below it, outside the line's 95 % prediction interval (±0.4).

What the dotted line is not. It extends the trend of 2000–2019 only to measure the gap. It is not a forecast: nothing in the data says that life expectancy will rejoin the line, or keep rising at that pace.

In 2023 the world was 1.1 years below its trend

World life expectancy at birth, 2000–2023, against the 2000–2019 trend



Our World in Data (UN WPP 2024 and others), 2000–2023 · make_report.py · git 4c16514

Figure 2. Source: UN World Population Prospects (2024) and others, via Our World in Data (CC BY 4.0); data for 2000–2023.

> **1.1 years**
> separate the world in 2023 from the line it followed for twenty years. A return to 2019 is not a return to the trend.

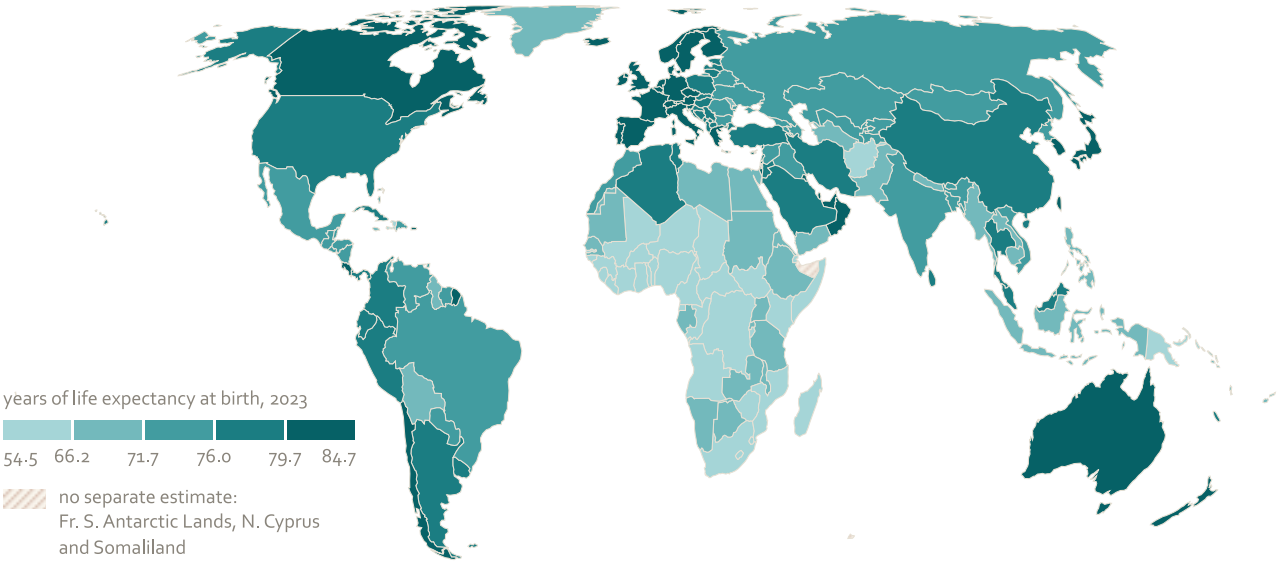
3 The gap

Where a child is born matters more than the pandemic did. In 2023 life expectancy ranged from 54.5 years in Nigeria to 84.7 in Japan across the 173 countries on the map. The gap between high- and low-income countries, 16.5 years, is more than nine times the world's pandemic fall.

Reading the map. Each class holds a fifth of the countries. Small countries vanish at this scale; Figure 4 gives the 30 most populous, 78 % of all people.

Of the 35 countries with the lowest life expectancy, 31 are in Africa

Period life expectancy at birth in 2023, 173 countries in five classes of equal count (quintiles)

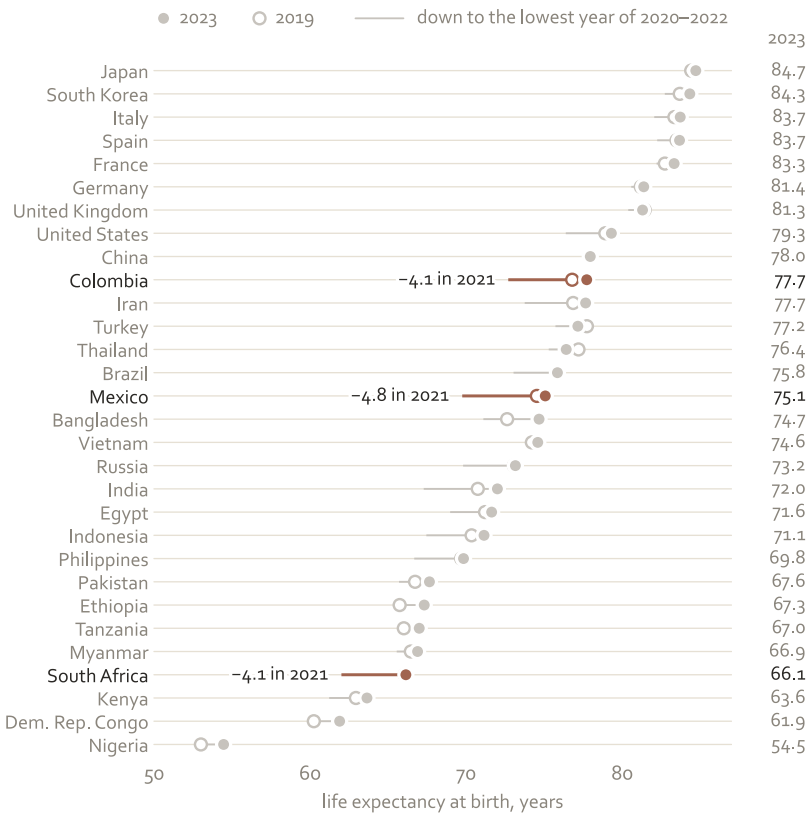


Our World in Data (UN WPP 2024 and others), 2023 · outlines Natural Earth v5.1.2, Equal Earth projection · make_report.py · git 4c16514

Figure 3. Boundaries are Natural Earth's and imply no judgement on the status of any territory. Sources: UN World Population Prospects (2024) and others, via Our World in Data (CC BY 4.0), data for 2023; Natural Earth v5.1.2.

Mexico, Colombia and South Africa fell four years or more and were back by 2023

Life expectancy at birth in the 30 most populous countries, 2019 to 2023



Our World in Data (UN WPP 2024 and others), Natural Earth v5.1.2 · make_report.py · git 4c16514

Figure 4. The 30 countries with the largest Natural Earth population estimates (2019). Sources: UN World Population Prospects (2024) and others, via Our World in Data (CC BY 4.0), data for 2019–2023; Natural Earth v5.1.2.

−4.8 years

in Mexico from 2019 to 2021, the deepest fall among the 30 most populous countries. By 2023 it was 0.5 years above 2019.

Sorted by the level of 2023, the figure shows the gap between countries down the page and the pandemic across each row. Three of the 30 had no year below 2019 in 2020–2022; three were still below 2019 in 2023.

Sources and method

Data. Period life expectancy at birth: UN World Population Prospects (2024), Human Mortality Database (2024), Zijdeman and Ribeira da Silva (2015), Riley (2005); processed by Our World in Data, downloaded 2026-10-12, CC BY 4.0. Outlines and population estimates for 2019: Natural Earth v5.1.2, public domain.

Method. The pandemic fall is the value of 2019 less the lowest of 2020–2022. The trend is a least-squares line through the world values of 2000–2019 (0.34 years a year) with its 95 % prediction interval. The map is in the Equal Earth projection (Šavrič, Patterson and Jenny 2018), without Antarctica.

Reproduce.

```
python examples/python/report/
make_report.py
```

recomputes every number from the two data files, redraws the figures and prints these pages.

4 The options

The figures suggest three options. They are examples of the layout, not recommendations.

Measure against the trend

A return to the 2019 level (+0.6 years by 2023) reads as recovery. Against the trend of 2000–2019 the world is 1.1 years short. Reporting both keeps the distance in sight.

Look where the fall was deepest

From 2019 to the lowest pandemic year, lower-middle-income countries lost 2.3 years, upper-middle-income 1.5, high-income 1.2 and low-income 0.3.

Keep the long gap in view

High-income countries live 16.5 years longer than low-income ones, down from 30.6 in 1950: more than nine times the world's pandemic fall of 1.7 years.

5 The ask

An example of an ask. Publish world and income-group life expectancy every year beside the trend of 2000–2019, with the distance to it, until the 1.1-year gap closes.

This shows how the layout states an ask. It is an illustration, not advice to any organisation.